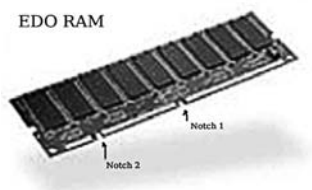


tests will use the same test settings. Press [T] for a detailed inspection of the memory module. This process consumes a lot of time, and you may have to leave your machine on through the night.



EDO RAM—Match the notches with those on the slots when fixing the RAM otherwise you may damage the module

If you have more than one RAM chip, then use one chip at a time while testing. This way, you'll know which one is faulty. Moreover, the problem could also arise due to a corroded interface on the motherboard. To counter this problem, perform the test using different combinations of slots and memory. Depending on the type of error and its gravity, you might need to call a technician to rectify the problem—which is most likely replacing the RAM chips or soldering new RAM slots to the motherboard.

8.1.2 Generic testing procedures

Keeping the system cache enabled when testing memory is the biggest mistake you can make, as most systems have a 'write-back cache', which invalidates the memory test. Data sent to main memory is first written into cache. A memory test program first writes data and then immediately reads it back. Eventually, the test program mainly reads and writes from the cache, and does not test all of the memory! The cache should therefore be disabled from the BIOS/CMOS setup before memory test programs are run.

The generic test procedure is a method by which a user can identify whether there is a problem, and then isolate it.

Procedure A: Detecting a problem

- i. Power on the system from an off state and observe the POST session. If no errors were detected, go to step ii otherwise follow procedure B.
- ii. Restart the system and enter the CMOS setup. Set all cache settings to 'Disabled'. Save and exit the CMOS setup and boot the system from a start-up disk containing a memory diagnostic tool.